Convergence of the Fourier Series

$$x_N(t) = \sum_{k=-N}^{N} a_k e^{j\omega_0 k t}$$

Let $e_N(t)$ denote the approximation error; that is

$$e_N(t) = x(t) - x_N(t) = x(t) - \sum_{k=-N}^{N} a_k e^{j\omega_0 k t}$$

In order to determine how good any particular approximation is, we need to specify a quantitative measure of the size of the approximation error. The criterion that we will use in the energy is the error over one period:

$$E_N = \int_T |e_N(t)|^2 dt$$

$$\therefore E_N = \int_T \left(x(t) - \sum_{k=-N}^{N} a_k e^{j\omega_0 k t} \right) \left(x^*(t) - \sum_{k=-N}^{N} a_k^* e^{-j\omega_0 k t} \right) dt$$

Let $a_i = b_i + j c_i$. Then

$$\frac{\partial E_N}{\partial b_i} = \frac{\partial}{\partial b_i} \left[\int_T x(t) x^*(t) dt - \int_T x(t) \sum_{k=-N}^{N} a_k^* e^{-j\omega_0 k t} dt - \int_T \left(\sum_{k=-N}^{N} a_k e^{j\omega_0 k t} \right) x^*(t) + \int_T \sum_{k=-N}^{N} \sum_{k=-N}^{N} dt \right]$$

$$\frac{\partial E_N}{\partial b_i} = 0 = -\int_T x(t)\, e^{-j\omega_0 i t}\, dt - \int_T e^{j\omega_0 i t}\, x^*(t)\, dt + 2b_i T \longrightarrow (1)$$

$$\frac{\partial E_N}{\partial c_i} = 0 = j\int_T x(t)\, e^{-j\omega_0 i t}\, dt - j\int_T e^{j\omega_0 i t}\, x^*(t)\, dt + 2c_i T \longrightarrow (2)$$

$$(1) + j(2) \Rightarrow 0 = -2\int_T x(t)\, e^{-j\omega_0 i t}\, dt + (2b_i + 2jc_i)T$$

$$\therefore T(b_i + jc_i) = \int_T x(t)\, e^{-j\omega_0 i t}\, dt = a_i T$$

$$\therefore \boxed{a_i = \frac{1}{T}\int_T x(t)\, e^{-j\omega_0 i t}\, dt}$$

If $x(t)$ has a fourier series representation, then the limit of E_N as $N \longrightarrow \infty$ is zero.

Now, the question is when does a periodic signal have a fourier series representation.

One class of periodic signal that are representable through the Fourier series is those signals which have finite energy over a single period, i.e.

$$\int_T |x(t)|^2 dt < \infty$$

When this condition is satisfied it is guaranteed that the integral below will result in finite a_k.

Further, let $x_N(t)$ be the approximation

$$x_N(t) = \sum_{k=-N}^{N} a_k e^{jk\omega_0 t}$$

Then we are guaranteed that the energy E_N converges to zero as more terms are added.

$$\therefore \quad e(t) = x(t) - \underbrace{\sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}}_{\hat{x}(t)}$$

then

$$\int_T |e(t)|^2 dt = 0.$$

This doesnot mean $\hat{x}(t)$ and $x(t)$ are the same at every point in time.

Dirichlet conditions:

<u>Condition 1:</u> (Bounded) Over any period, $x(t)$ must be absolutely integrable, i.e.,

$$\int_T |x(t)|\, dt < \infty$$

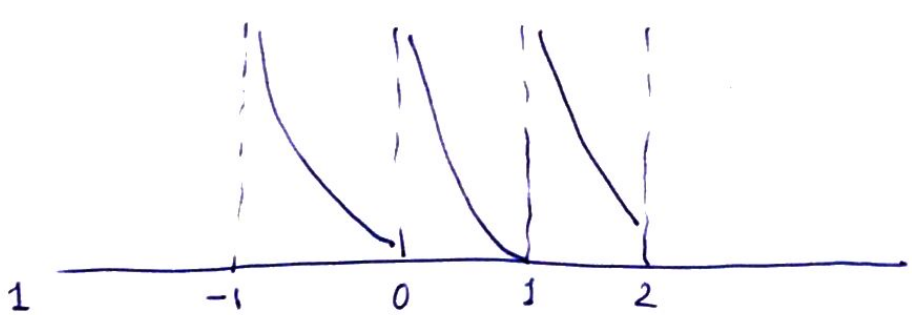
As with square integrability, this guarantees that each coefficients a_k will be finite, since

$$|a_k| \leqslant \frac{1}{T} \int_T |x(t) e^{-jk\omega_0 t}|\, dt = \frac{1}{T} \int_T |x(t)|\, dt$$

$$\therefore \int_T |x(t)|\, dt < \infty \Rightarrow |a_k| < \infty.$$

A periodic signal that violates the first Dirichlet condition is

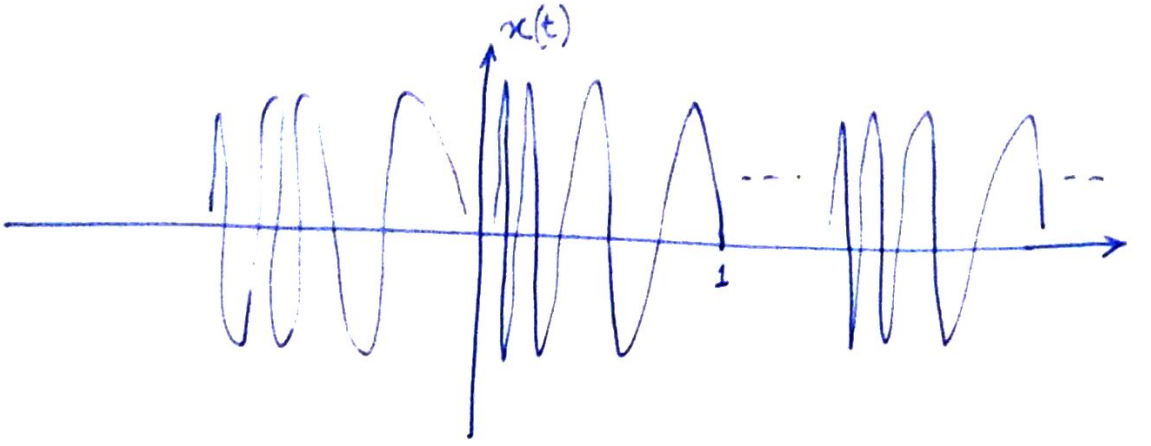
$$x(t) = 1/t \quad ; \quad 0 < t \leqslant 1$$



Condition 2: In any finite interval of time, $x(t)$ is of bounded variation; that is there are finite number of maxima and minima in one period.

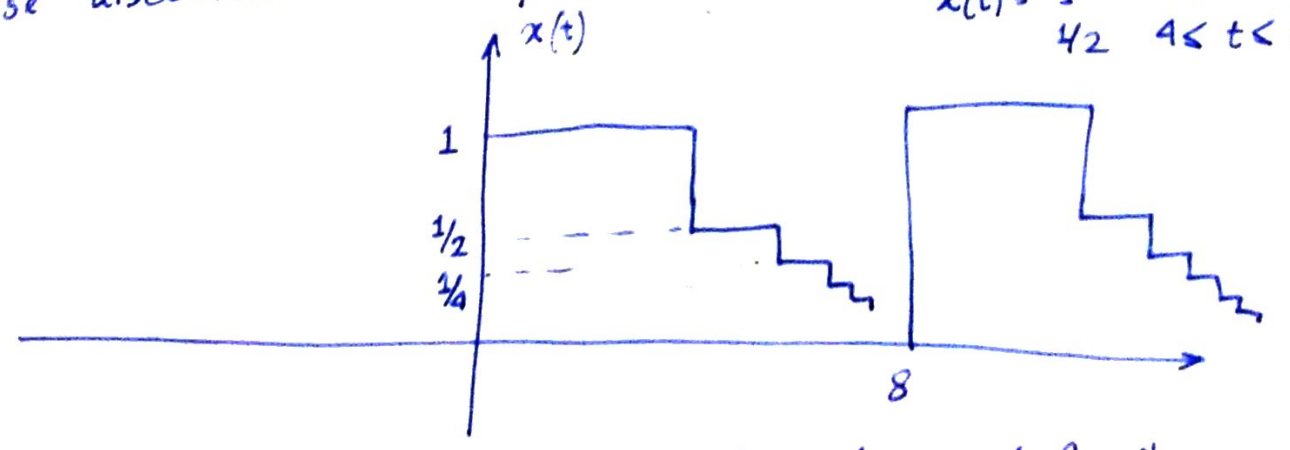
A signal that meets Condition 1 but not Condition 2.

$$x(t) = \sin\left(\frac{2\pi}{t}\right) \qquad 0 < t \leq 1$$



Condition 3: In any finite interval of time, there are only finite number of discontinuities. Further, each of these discontinuities are finite.

$$x(t) = \begin{cases} 1 & 0 \leq t \leq 4 \\ 1/2 & 4 \leq t < 6 \end{cases}$$



The value of $x(t)$ decreases by a factor of 2 whenever the distance from t to 8 decreases by a factor of 2

→ For a periodic signal that has no discontinuities, the Fourier series converges and equals the original signal at every value of t.

→ For a periodic signal with finite number of discontinuities in each period, the Fourier series equals the signal at every point except at the isolated points of discontinuity. At the isolated points of discontinuity, the series converges to the average value of the signal on either side of the discontinuity.

In this case, the difference between the original signal and its Fourier series represention contains no energy, and hence the two signals can be thought off as the same. for all practical purposes.

Dirichlet's theorem says in particular that under Dirichlet's condition the fourier series for f converges and is equal to f wherever f is continuous

$$f(x) = x^n \sin(x^{-1}) \text{ on } [-\pi, \pi] \qquad n \geq 3$$